

Requirement Analysis

Solution Requirements (Functional & Non-functional)

Date	04 September 2026
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Project Name	Metro Ticket Generating System
Maximum Marks	

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement
FR-1	Application & Data Mode	Custom Metro application and Metro Station Details table.
FR-2	Station Management	Store and manage available metro station names
FR-3	Ticket Booking	Capture Starting From, Going To, Journey Type and Passenger Count
FR-4	Fare Calculation	Automatically calculate fare based on route, journey type and passenger count
FR-5	Payment Details	Provide payment mode selection and additional payment-mode input when required
FR-6	Catalog Item	Book a Metro Ticket catalog item with configured variables
FR-7	Portal Interface	Provide the ticket-booking experience through ServiceNow Service Portal
FR-8	Client-side Automation	Dynamically populate fare values based on passenger and journey selections
FR-9	QR Ticket Generation	Generate a digital QR code after ticket submission
FR-10	Data Validation & Testing	Validate form behavior, ticket creation, fare calculation and QR generation

Non-functional Requirements

NFR	Requirement	Description
NFR-1	Usability	Simple and guided ticket-booking interface
NFR-2	Security	Restrict direct access to backend ticket data using ServiceNow access controls
NFR-3	Reliability	Fare and field rules should behave consistently
NFR-4	Performance	Ticket form and fare calculation should respond within normal ServiceNow interaction time
NFR-5	Availability	System availability depends on the ServiceNow instance
NFR-6	Scalability	Design should support future payment, notification and analytics enhancements

Solution Requirements — Functional Coverage

FR-1	Application setup	Custom Metro Ticket application
FR-2	Station data	Metro Station Details table with station names
FR-3	Ticket booking	Starting From, Going To, Type of Journey, No. of Passengers
FR-4	Fare calculation	Fare calculated automatically based on route and journey type
FR-5	Payment	UPI, Card, Others and conditional payment-mode field
FR-6	Portal	ServiceNow Service Portal ticket-booking interface
FR-7	QR ticket	Digital QR ticket generated after submission
FR-8	Validation & testing	Validate form behavior, fare calculation, record creation and QR generation

Figure 1: Functional requirement coverage.